N83 full-D18 third input and execution preflight

2026-10-26, primary curve: `icv1-f2m83-tm6151469093347-debefd74`, Koblitz `a=0` over GF(2^83), prime subgroup order `241785163923079621668568 9`.

Observed state: `INPUT_CUSTODY_PASS`; `LOCAL_CAPACITY_GATE_FAIL`; `CLOUD_DIRECT_READ_DENIED`; `ON_DEMAND_SMOKE_LAUNCH_DENIED`; `WORKER_NOT_LAUNCHED`. This is a third fresh input check for the frozen native positive-then-natural protocol and assignment amendment. The access and launch diagram shows both checked routes. It is a process diagram because neither solver arm ran; there are no comparable algorithm timings. A later infrastructure smoke also failed before instance allocation. `selected_best_total_runtime` remains `null`.

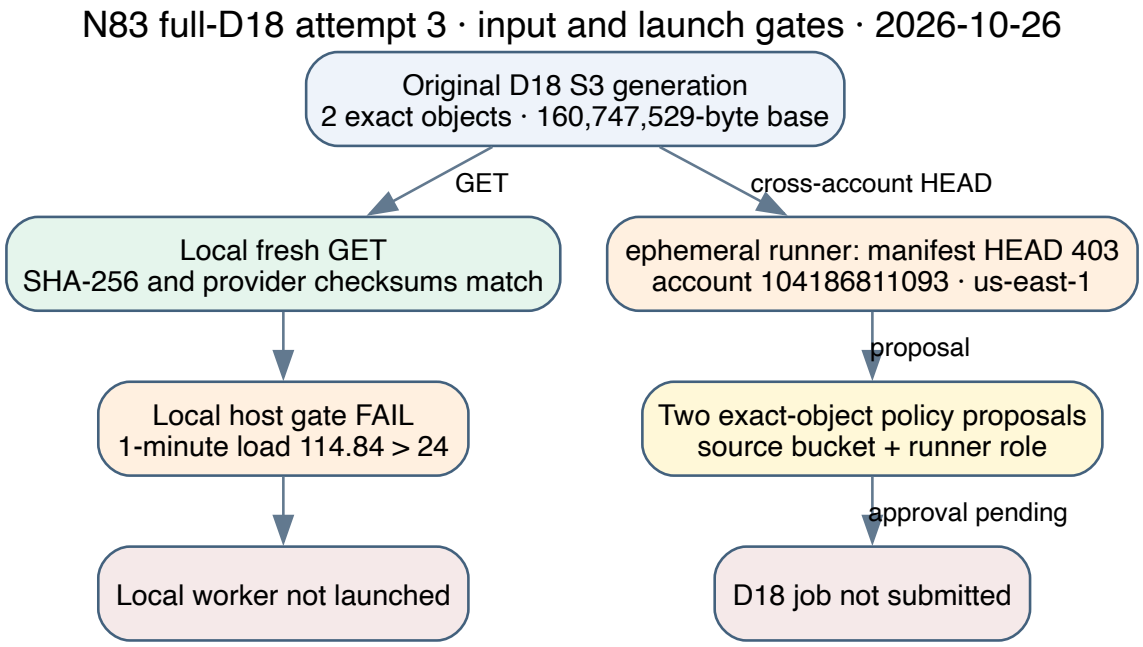


Figure 1: Input custody and execution gates

Source and input identity

The clean sparse source checkout was `45ce91d503ee0d0acc57e3d8d5cb49108c90ded6` from PR #2649. Its `git status --porcelain` was empty. The release worker and separate verifier, built earlier from this exact commit, retained SHA-256 respectively `11220d8b838bf5cac2985dbcba4ea96e69a64b70dc54044d180cf6a3e8970e49` and `e89578703888568d67f79e5b2645e47b911a30a9d7a8a3d57e795be997296b71`. Neither binary was launched in this preflight.

Fresh checksum-enabled `aws s3api get-object` calls to the original `crypto-autoresearcher` bucket fetched the producer-bound custody wrapper and the full D18 compressed object. SHA-256 of the committed producer manifest, fresh S3 wrapper, and local compressed object matched, in order:

`1d3650eabc4662d8f2597498d7ac92b7cd0a4134ac5f8155aaa49c7af3407dce`
`1b93351febd47dac4de8c5dd39dbf58564d0057a044054baf448d978f6c0d616`
`eaef94ef45b3d2d4cd2ec2f40c1173b4cdb2a437b7e618a858c73fb1caf3902b`

The wrapper and object provider responses both report `ChecksumType=FULL_OBJECT`; the object response reports `ContentLength=160747529`. The compressed object remains at the immutable S3 key in AWS_RESUME.md; it is not copied into Git. These checks establish the exact fetched bytes, not relation yield or solver correctness.

Execution gates

The local host preflight at `2026-10-26T21:40:48Z` recorded one-minute load **114.84**, versus the frozen launch maximum **24**. The scratch filesystem reported 40 GiB free, above the 2 GiB minimum. Because load failed, the local worker was not launched; this single observation does not measure its potential runtime. The swapout counter in the receipt is cumulative and is not interpreted as a run delta because no arm ran.

The repository's deployed `ephemeral` runner configuration was accessible through the configured `ecc-challenges-new` AWS profile in account `104186811093`, `us-east-1`; its instance profile is `ephemeral-runner`. A `HeadObject` for the original D18 manifest through that profile returned `403` (exit, stderr). Read-only IAM inspection found no exact D18 read in the runner role, and the source bucket had no bucket policy. The two-sided exact-read proposal records the minimal source-bucket and runner-role policies for a direct original-object GET. **Neither proposed policy was applied and no hosted D18 job was submitted.** The separate previously approved `rho-gh-snapshot-readonly-20261024` user grant remains a different access path.

The new opt-in `--on-demand` runner path passed the Docker-free fake-AWS market test and eight Rust CLI unit tests. A real `--no-source --toolchain none --on-demand --type t3.small --ttl 20m` market smoke then created job receipt `20261026-215459-n83-ondemand-market-smok-7d2a65`, with `spot:false`, but its status is `error` at `RunInstances`. The CLI error named an explicit deny in IAM policy `SpotOnlyEC2Launch` version `v2`, whose first two statements deny non-Spot instance launches outside their stated Auto Scaling or recovery-template conditions. A tag-specific instance readback returned `[]`: no EC2 instance was allocated. This attempted smoke did not fetch D18 input or execute cryptographic code. The exact S3 read proposals and a separately authorized non-Spot launch path are both needed for the current hosted plan.

Outcome and next action

Requested step	Status in attempt 3
Exact S3 input custody	Verified for the fresh local GET, hashes above
Constructed cadence-32 SAT and separate assignment replay	Not launched; local host gate failed
One public natural query	Not launched; positive gate not passed
Hosted direct original S3 fetch	`403` in deployed runner account; exact grant proposed
On-Demand runner smoke	`RunInstances` explicit deny; no instance allocated
Cold total-runtime selection	Unmeasured; no factor-base winner selected

The next executable action is either a local host that passes the frozen load and swap gates, or an authorized deployed runner that can directly GET the two original S3 objects and meets the frozen host and checkpoint/isolation requirements. Run one constructed control at its 300-second internal cap, require separate group and retained-assignment replay if it reaches SAT, and only then run the one 2,400-second public natural query. Preserve any cap or infrastructure failure. Natural yield, closure, useful rank, logs, unseen-target descent, fully charged cold cost, matched rho and finite-grid coverage remain open. No new algorithm result changes the canonical performance or boundary graphs.